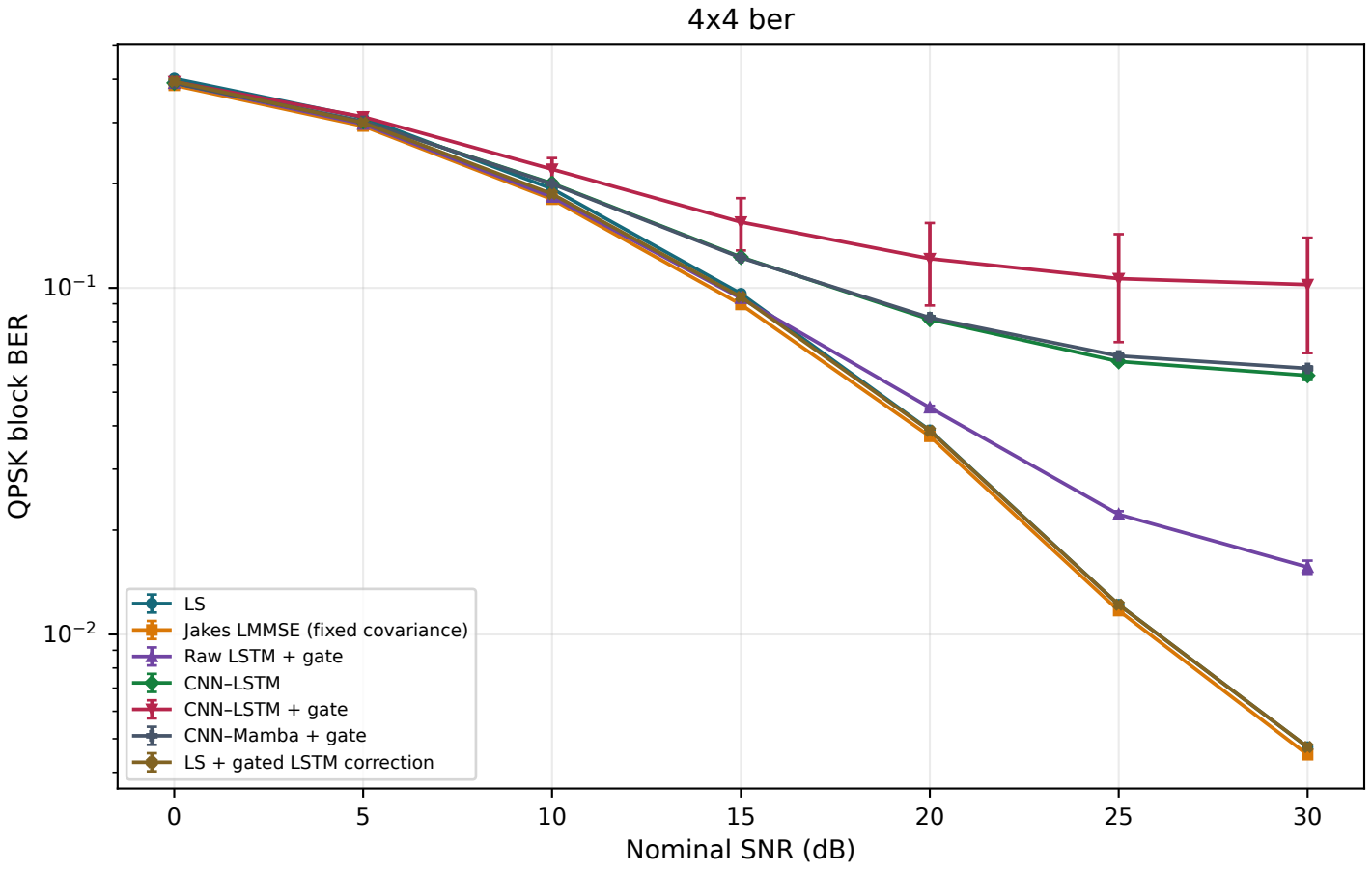
4x4 ber
QPSK block BER
10^{-1}
10^{-2}
0
5
10
15
20
25
30
Nominal SNR (dB)
LS
Jakes LMMSE (fixed covariance)
Raw LSTM + gate
CNN–LSTM
CNN–LSTM + gate
CNN–Mamba + gate
LS + gated LSTM correction